CARBON QUANTUM DOTS FOR SOLAR CELL APPLICATIONS

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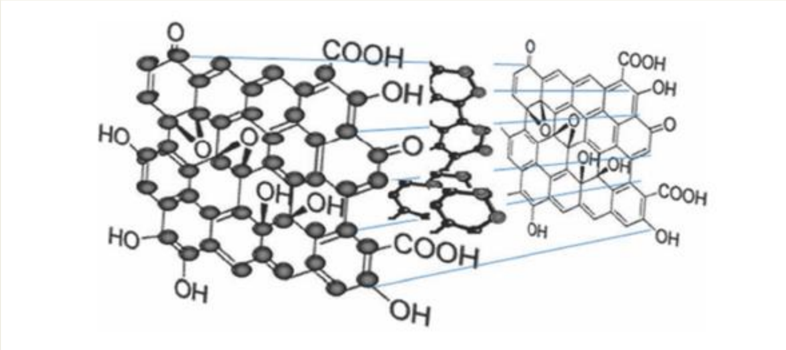
CHE 494
May 7th , 2020

Introduction



- The unique properties of carbon-based nanomaterials such as carbon nanotubes, graphene sheets and fluorescent carbon nanoparticles, or carbon quantum dots (CQD) have motivated wide-ranging studies due to their great potential for a wide variety of technical applications.
- CQDs possess the intrinsic advantages of being an abundant element and having relative low toxicity.
- The general description of CQDs is that of nanometer size particles consisting of sp2 hybridized graphitic core functionalized with polar carboxyl or hydroxyl groups on the surface.

Importance of CQD’s

- Compared to traditional semic … carbon-based quantum dots are super … tness, facile modification and high re…
- The superior biological propert … good biocom- patibility, entrust them … biomolecule/drug delivery.
- The outstanding electronic pro… d acceptors, causing chem- iluminescence and electrochemical luminescence, endow them with wide potentials in optronics, catalysis and sensors.

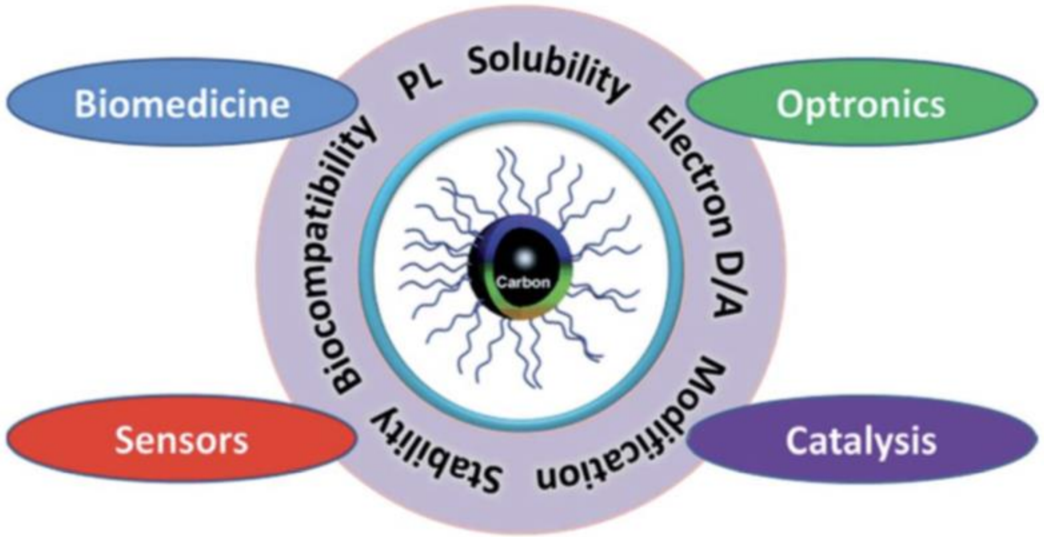


Fig. 1 CQDs with unique properties have great potential in biomedi-cine, optronics, catalysis and sensors.

Goal of the Research

- Carbon quantum dots: synthesis, properties and applications
  - *Journal of Materials Chemistry C*
  - *Authors:Youfu Wang and Aiguo Hu*
- Describe the recent progress in the field of CQDs, focusing on their synthetic methods, size control, modification strategies, optical properties, luminescent mechanism, and applications in biomedicine, optronics, catalysis and sensor issues
- Discuss the main methods for CQDs synthesis, the size control via confined pyrolysis and the modification of CQDs, including functionalization, doping and nano-hybrids.

Synthesis, Size Control and Modification

- Several development procedures have been developed to prepare and improve CQDs, which can be loosely divided into "Top-down" and "Bottom-up" solutions
- Both can be changed during a pretreatment, prep phase or a post-treatment phase. Three problems in the preparation of CQDs need to be noted:
  - *carbonaceous accumulation during carbonization, which can be prevented utilizing electrochemical synthesis, continuous pyrolysis or solution chemistry methods,*
  - *size regulation and uniformity, which is critical for uniform properties and mechanistic analysis, and can be improved by post-treatment, such as gel electrophoresis, centrifugation and*
  - *surface properties that are critical for solubility and selected applications*

Chemical Ablation

- Strong oxidizing acids carbonize small organic molecules to carbon-rich materials. These materials can be further cut into small sheets through controlled oxidation.
- Peng and Travas-Sejdic reported a simple route to prepare luminescent CQDs in an aqueous solution by dehydrating carbohydrates with concentrated $H2SO4$, followed by breaking the carbonaceous materials into individual CQDs with $HNO3$
- However, this method is associated with harsh conditions and severe processes.
- The emission wavelength of these CQDs can be modified by differing the starting material and the length of a nitric acid treatment. The multicolor emission capabilities and nontoxic nature of the CQDs produced allow them to be applied in life science research.

Electrochemical Carbonization

- The preparation of CQDs from electrochemical soaking is a powerful method in production. The process uses various carbon materials in bulk as precursors.
- The electrochemical method involves a top-down solution, where the chemical cutting process of a carbon material, such as carbon nanotubes or graphite, is used
- Carbon dots prepared through contemporary electrochemical methods have varying sizes and would require further separation through filtration or chromatography to obtain uniformity

Microwave Irradiation

- The use of microwave irradiation is a low cost and rapid method in the synthesis of CQDs.
- Using sucrose as the carbon source and diethylene glycol (DEG) as the reaction media, green luminescent CQDs were obtained within one minute under microwave irradiation.
- Sucrose is used as the carbon source with diethylene glycol (DEG) as media for reaction. Through one minute of microwave radiation, green luminescent CQDs were obtained. The resultant CQDS are highly bio-compatible and have increased potential for applications in the biomedical field.

Optronics

- The CL of CQDs opens up new opportunities for their potential in the determination of reductive substances. The dual role of CQDs as an electron donor and acceptor offers great potential in optronics and catalysis
- Research focus on two main solar cell applications
  - *Dye-sensitized solar cells (DSCs)*
  - *Organic solar cells (OSCs)*

Dye-sensitized Solar Cells (DSCs)

- DSCs have arous … processing.
- CQDs with stable … its potential in DSCs.
- Inspired by natur… emiconductor complex system for the fal… s.
- CQDs not only en… ons due to the synergistic hyper… ectively suppressed the recombination of photogenerated electrons, thereby leading to a significantly enhanced photoelectric conversion efficiency

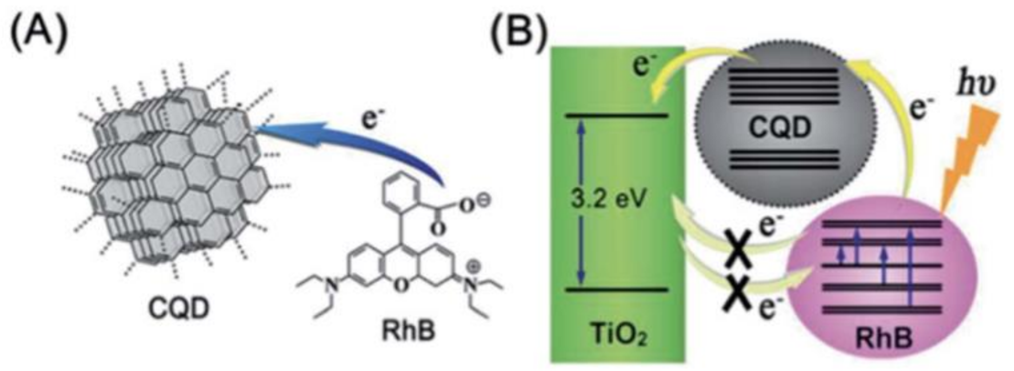


Fig. 25 (A) Scheme of electron transfer from RhB to CQDs. (B) CQDs serve as an electron transfer intermediary for bridging the photo-generated electrons and suppressing their recombination. Adapted with permission.[117] Copyright 2013, American Chemical Society.

Organic Solar Cells

- A simple and effective method to prep
  was developed
- The P3HT:PCBM based solar cell harv
  covering a part of visible light (380–7
  irradiance output
- Power conversion efficiency of the latt
  layer of the composite on the cover gl
  near ultraviolet and blue-violet portior

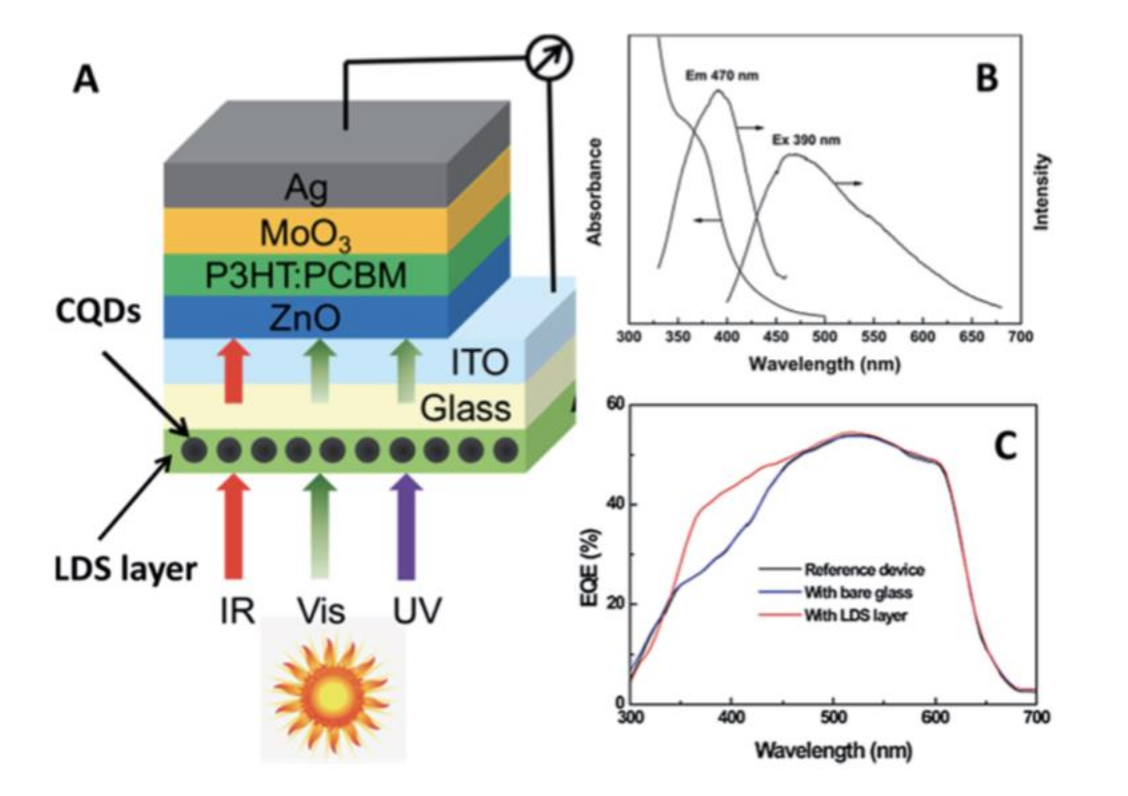


Fig. 26 (A) Schematic architecture of the BHJ solar cell with LDS layer. (B) UV-vis absorption and PL spectra of the CQDs filled polysiloxane composite (~20 μm thick) coated on glass. (C) Wavelength dependences of external quantum efficiency of the P3HT:PCBM-based solar cell with and without the LDS layer. Adapted with permission.[71] Copyright 2014, Elsevier.

Challenges

- Complicated methods for their isolation, purification, and functionalization, their generally low quantum yields, and complexity in their configuration, design, and structure are some of the problems that need to be resolved before they can fully outperform their quantum dot equivalents for semiconductors.
- Quantum dots may also have surface defects that can impact the electrons and holes recombination by serving as temporary "traps."

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